

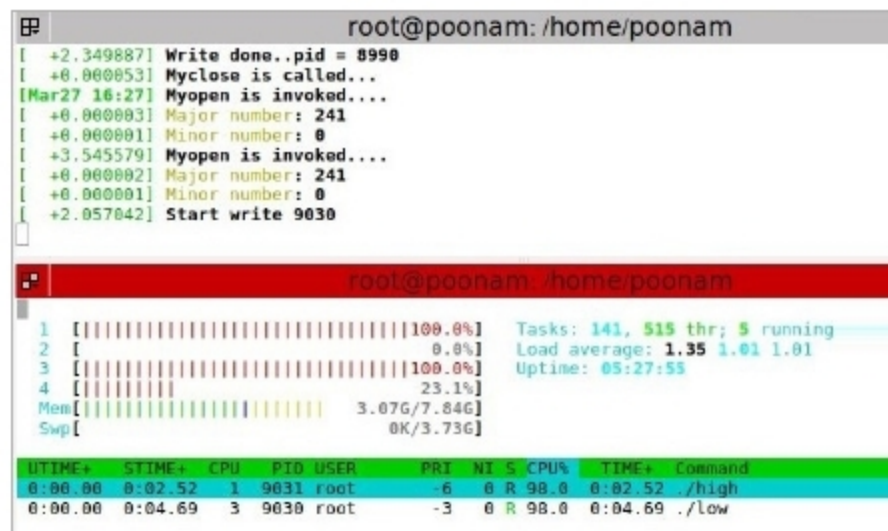


Figure 9: Both processes execute simultaneously

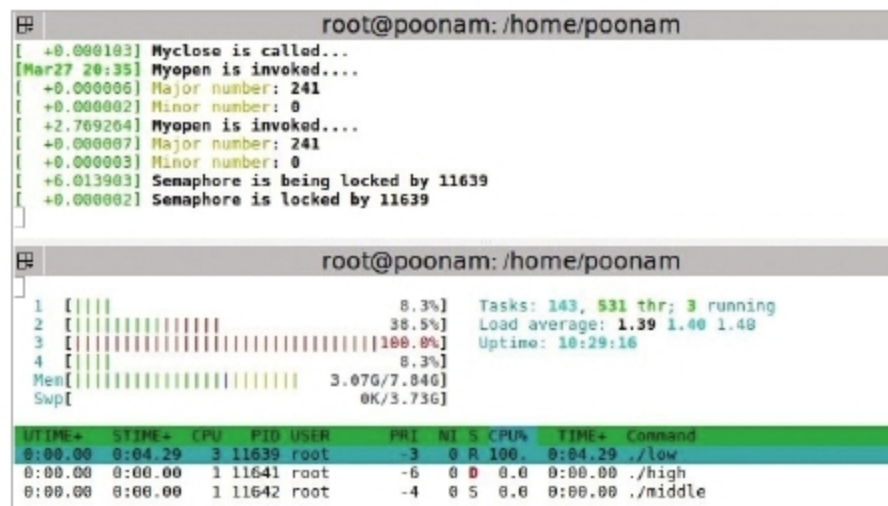


Figure 10: High waits for low to complete the execution of CS

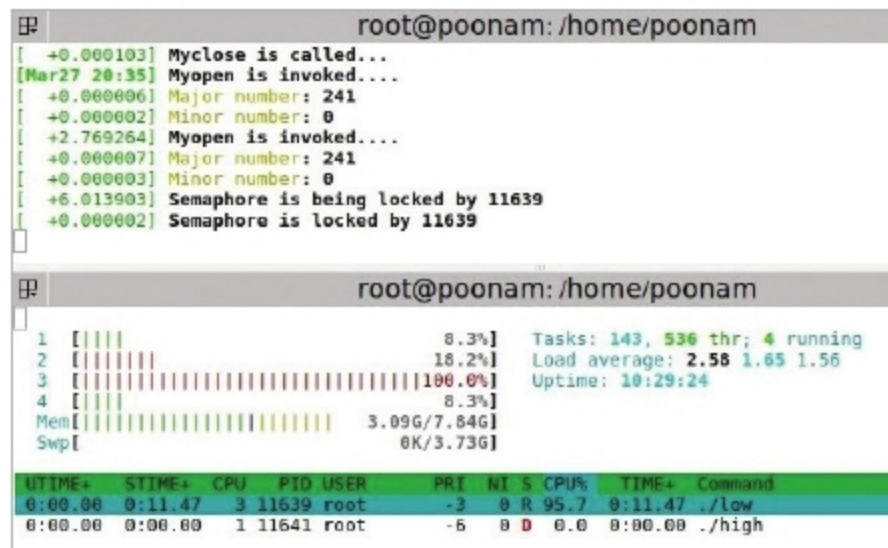


Figure 11: Low resumes the execution in CS

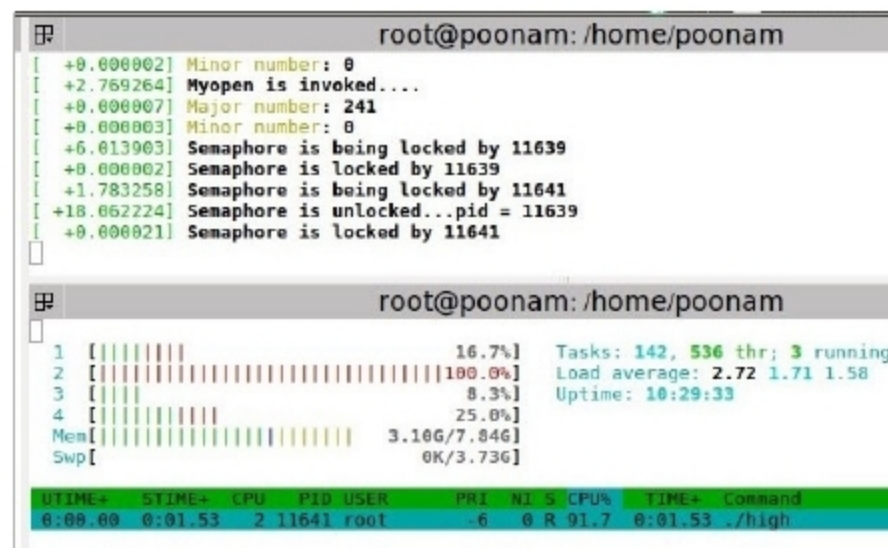


Figure 12: High executes in CS

Observation: When the kernel is PRE-EMPTIVE, that means, at the kernel level, pre-emptions are allowed except for critical sections. Here, the low priority process is running in the critical section, and the high priority and middle priority processes come in. High doesn't pre-empt the low priority process but rather, it waits for the lower priority process to complete executing the critical section. The middle priority process runs in between.

Note that the priorities of processes remain unchanged.

RTMutex

The setup: A low priority process is executed first. Then a high priority process comes in which also wants to write to the same buffer. A middle priority process comes in which does not want to access the device driver. Middle is a simple application program.

Observation: The low priority process is running in critical section, and the high and middle priority processes come in. As soon as the high priority process comes in, the priority of the low priority process gets increased to a high priority value and remains the same till the process completes executing the critical section. So, high doesn't pre-empt the low priority process. After completing the low priority process, like in the case of semaphores, the middle process can't execute as a higher priority process is available.

Observe that the priority of low priority process updates to 5 when in the critical section, and returns to 2 after execution.

Multiprocessor systems

Without lock setup: A low priority process is executed first. Then a high priority process comes in, which also wants to write to the same buffer.

Observation: When both processes are running on different processors, both execute in parallel. But the buffer update problem remains the same. At this point, we can't make out which process has updated the buffer.

Semaphore

The setup: A low priority process is executed first. Then a high priority process comes in, which also wants to write to the same buffer. A middle priority process comes in, which does not want to access the device driver. Middle is a simple application program.

Observations: Here, all processes are running on different processors. So, the middle priority process has no dependency on the other two processes. For the other two processes, one waits for the other to complete executing the critical section.

rt-mutex

The setup: The low priority process is executed first. Then the high priority process comes in, which also wants to write in the same buffer. Then the middle priority process comes in and it does not want to access the device driver. Middle is a simple application program.

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